# 3.1 Three Views of a Linear System

Chapter 1 kept running into the same kind of question. Is this vector in the span of those? Is this list independent? Every time, the question turned into a handful of equations in unknown coefficients, and we solved them by hand. This chapter gives those equations a name, looks at them from three sides, and turns solving them into an algorithm with a proof behind it. This first section sets up the language and answers a structural question before any computing: what can the set of all solutions look like?

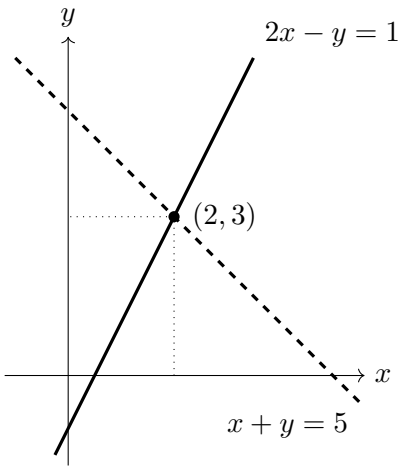
## 3.1.1 One system, three pictures

Here is a system of two equations in two unknowns, over $\mathbb{R}$:

$$\begin{aligned} 2x - y &= 1, \\ x + y &= 5. \end{aligned}$$

Adding the equations gives $3x = 6$, so $x = 2$, and then $y = 5 - x = 3$. We can read this one small system in three different ways, and each will be useful.

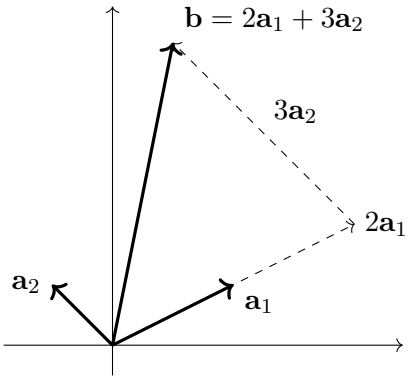
**The row picture.** Each equation describes a line in the plane: $y = 2x - 1$ and $y = 5 - x$. A solution is a point on **both** lines, so solving the system means intersecting them. The lines meet at $(2, 3)$.




**The column picture.** Stack the coefficients of $x$ into one column and those of $y$ into another. The system says

$$x \begin{pmatrix} 2 \\ 1 \end{pmatrix} + y \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}.$$

Now the unknowns are **weights**. We are asked how much of the vector $\mathbf{a}_1 = (2, 1)$ and how much of $\mathbf{a}_2 = (-1, 1)$ to combine to reach $\mathbf{b} = (1, 5)$. The answer $x = 2$, $y = 3$ says: walk twice along $\mathbf{a}_1$, then three times along $\mathbf{a}_2$.




**The matrix picture.** Collect the coefficients into a matrix and the unknowns into a column. By the definition of matrix multiplication (Definition 0.9.9), the system is the single equation

$$\begin{pmatrix} 2 & -1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 1 \\ 5 \end{pmatrix}, \qquad \text{that is,} \qquad \mathbf{Ax} = \mathbf{b}.$$

The three pictures say the same thing, and they are tied together by one fact from Chapter 0: $\mathbf{Ax} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$, a combination of the columns of $\mathbf{A}$ weighted by the entries of $\mathbf{x}$ (Theorem 0.9.3). The row picture is the one we draw. The column picture connects to Chapter 1, since it asks whether $\mathbf{b}$ lies in a span. The matrix picture is the one we compute and prove things with.

*Remark.* With more unknowns the pictures can no longer be drawn, but they do not change. An equation in three unknowns whose coefficients are not all zero describes a plane in $\mathbb{R}^3$, and a system asks where several planes meet. In the column picture, three unknowns mean three columns, and we ask whether a combination of them hits $\mathbf{b}$.

## 3.1.2 Linear systems

The example had two equations and two unknowns over $\mathbb{R}$. The general notion allows any numbers of each, and any field. Allowing any field is not generality for its own sake. Systems over $\mathbb{F}_2$ describe parity checks and switching puzzles, and systems over $\mathbb{Q}$ are what a computer can solve exactly.

*A linear system is a finite list of equations, each saying that a fixed combination of the unknowns equals a fixed scalar.*

**Definition 3.1.1. Linear system**

Let $F$ be a field and let $m, n \geq 1$. A **system of $m$ linear equations in $n$ unknowns $x_1, \ldots, x_n$ over $F$** is a list of equations

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1, \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2, \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m, \end{aligned}$$

where all $a_{ij}$ and $b_i$ lie in $F$.

- The **coefficient matrix** is $\mathbf{A} = (a_{ij}) \in M_{m\times n}(F)$, the **right-hand side** is $\mathbf{b} = (b_1, \dots, b_m) \in F^m$, and the system is written $\mathbf{Ax} = \mathbf{b}$.
- The **augmented matrix** is the $m \times (n+1)$ matrix $[\,\mathbf{A} \mid \mathbf{b}\,]$ obtained by appending $\mathbf{b}$ to $\mathbf{A}$ as a last column.
- A **solution** is a vector $\mathbf{s} = (s_1, \dots, s_n) \in F^n$ such that **every** equation holds when $x_j = s_j$ for all $j$; equivalently, $\mathbf{As} = \mathbf{b}$. The **solution set** is the set of all solutions, a subset of $F^n$.
- The system is **consistent** if it has **at least one** solution, and **inconsistent** if its solution set is empty.

In words: the coefficient matrix records the left-hand sides, row $i$ for equation $i$ and column $j$ for unknown $x_j$. The augmented matrix records the whole system, and it is the object we will manipulate in the next section. A solution is a single vector of $F^n$, and it must satisfy all $m$ equations at once, not just some of them. "Equivalently, $\mathbf{As} = \mathbf{b}$" holds because the $i$-th entry of $\mathbf{As}$ is $a_{i1}s_1 + \cdots + a_{in}s_n$ by Definition 0.9.9, and two columns are equal exactly when all their entries agree.

Two small words deserve attention. A solution lives in $F^n$, with entries in **the field** $F$: the equation $2x = 1$ has the solution $\frac{1}{2}$ over $\mathbb{Q}$, but over $\mathbb{F}_2$ it reads $0 = 1$ and has none. And "consistent" means **at least one** solution, not exactly one.

**Example. None, one, or a whole line**

Find the solution set of each system over $\mathbb{R}$, and describe it in the row picture.

(a) $x + 2y = 1$, $2x + 4y = 5$.

(b) $2x - y = 1$, $x + y = 5$.

(c) $x + 2y = 1$, $2x + 4y = 2$.

*Solution.*

(a) Suppose $(x, y)$ is a solution. Doubling the first equation gives $2x + 4y = 2$, while the second says $2x + 4y = 5$. Then $2 = 5$, which is false. So there is no solution, and the system is **inconsistent**. In the row picture, the lines $y = \frac{1}{2} - \frac{1}{2}x$ and $y = \frac{5}{4} - \frac{1}{2}x$ have the same slope and different intercepts: they are parallel and never meet.

(b) As computed at the start of the section, any solution has $x = 2$ and $y = 3$; conversely $2 \cdot 2 - 3 = 1$ and $2 + 3 = 5$. The solution set is $\{(2, 3)\}$, the single point where two nonparallel lines cross.

(c) The second equation is twice the first, so a pair $(x, y)$ satisfies both exactly when it satisfies $x + 2y = 1$, that is, when $x = 1 - 2y$. Writing $t$ for $y$, the solution set is $\{(1 - 2t, t) : t \in \mathbb{R}\}$. Both equations describe the **same** line, and every point on it is a solution.

The three answers are the three things two lines can do: miss, cross, or coincide. Over $\mathbb{R}$, and over any infinite field, no other sizes are possible: once we know the shape of the solution set (Theorem 3.1.3 below), this is a short argument, which you will give in Exercise (C1: How many solutions?). For instance, no system over $\mathbb{R}$ has exactly two solutions. Over a finite field the picture is different.

**Example. Exactly two solutions over $\mathbb{F}_2$**

Find all solutions of the system $x + y + z = 1$, $y + z = 0$ over $\mathbb{F}_2$. Then compare with the same system over $\mathbb{R}$.

*Solution.* In $\mathbb{F}_2$ we have $-1 = 1$, so the second equation says $y = -z = z$. Substituting into the first gives $x + 2z = 1$, and $2z = 0$ in $\mathbb{F}_2$, so $x = 1$. Conversely, every $(1, z, z)$ satisfies both equations: $1+z+z = 1+2z = 1$ and $z+z = 0$. Since $z$ can only be 0 or 1, the solution set is

$$\{(1, 0, 0),\ (1, 1, 1)\},$$

with exactly two elements.

Over $\mathbb{R}$ the second equation gives $y = -z$, and then the first gives $x = 1$. The solution set is $\{(1, -t, t) : t \in \mathbb{R}\}$, a line in $\mathbb{R}^3$ with infinitely many points. The equations are the same, but the field decides how many values the free unknown can take.

Now a nonexample, by a minimal change. The equation $x+y^2 = 1$ looks like one of the equations above, and it still has a left side built from the unknowns and a constant on the right. But the unknown $y$ appears squared, so the left side is not of the form $a_1x_1 + a_2x_2$ with **fixed scalars** $a_1, a_2$. The clause that fails is "a fixed combination of the unknowns". Its solution set in $\mathbb{R}^2$ is a parabola, not a line, and nothing in this chapter applies to it.

*Warning.* **"Consistent" does not mean "has exactly one solution".** The system $x + 2y = 1$, $2x + 4y = 2$ of Example (None, one, or a whole line) (c) is consistent, and it has infinitely many solutions. Consistency only rules out the empty solution set. Whether the solution is unique is a separate question, answered by Corollary 3.1.4 below.

Why record the augmented matrix at all, when $\mathbf{Ax} = \mathbf{b}$ is shorter? Because solving a system never touches the names of the unknowns. Adding two equations adds their coefficients and their right-hand sides, and nothing else. The augmented matrix keeps exactly this data and throws away the symbols $x_1, \ldots, x_n$ and the plus signs. In the next section every manipulation is performed on it.

One kind of system is always consistent, because it has an obvious solution.

**Definition 3.1.2. Homogeneous system**

A linear system $\mathbf{Ax} = \mathbf{b}$ over $F$ is **homogeneous** if $\mathbf{b} = \mathbf{0}$. The vector $\mathbf{0} \in F^n$ is a solution of every homogeneous system $\mathbf{Ax} = \mathbf{0}$, called the **trivial solution**. For any system $\mathbf{Ax} = \mathbf{b}$, the system $\mathbf{Ax} = \mathbf{0}$, with the same coefficient matrix, is its **associated homogeneous system**.

The trivial solution exists by the zero rule of Theorem 0.9.4: $\mathbf{A0} = \mathbf{0}$. So for a homogeneous system the interesting question is never "is there a solution?" but "is there a **nontrivial** one?". For example, $x + 2y = 0$, $2x + 4y = 0$ has the nontrivial solution $(-2, 1)$, while $2x - y = 0$, $x + y = 0$ has only the trivial one: adding the equations gives $3x = 0$, so $x = 0$, and then $y = -x = 0$.

*Quick check.* Over $\mathbb{R}$, write the system $x_1 - x_2 + 3x_3 = 4$, $2x_2 + x_3 = 0$ in the matrix picture and the column picture, and give its augmented matrix. Is $(4, 0, 0)$ a solution? Is $(1, 0, 1)$?

*Solution.* With $\mathbf{A} = \begin{pmatrix} 1 & -1 & 3 \\ 0 & 2 & 1 \end{pmatrix}$ and $\mathbf{b} = (4, 0)$, the system is $\mathbf{Ax} = \mathbf{b}$. In the column picture it reads $x_1(1, 0) + x_2(-1, 2) + x_3(3, 1) = (4, 0)$. The augmented matrix is

$$[\,\mathbf{A} \mid \mathbf{b}\,] = \left[\begin{array}{ccc|c} 1 & -1 & 3 & 4 \\ 0 & 2 & 1 & 0 \end{array}\right].$$

For $(4, 0, 0)$: $4 - 0 + 0 = 4$ and $0 + 0 = 0$, so it is a solution. For $(1, 0, 1)$: the first equation gives $1 - 0 + 3 = 4$, but the second gives $0 + 1 = 1 \neq 0$. One failing equation is enough, so it is not a solution.

## 3.1.3 Consistency is a question about span

The column picture turns "does this system have a solution?" into a question from Chapter 1.

**Theorem 3.1.1. Consistency and the span of the columns**

Let $\mathbf{A} \in M_{m\times n}(F)$ have columns $\mathbf{a}_1, \ldots, \mathbf{a}_n \in F^m$, and let $\mathbf{b} \in F^m$. Then the system $\mathbf{Ax} = \mathbf{b}$ is consistent **if and only if** $\mathbf{b} \in \operatorname{span}(\mathbf{a}_1, \ldots, \mathbf{a}_n)$. More precisely, $\mathbf{s} = (s_1, \ldots, s_n)$ is a solution exactly when $\mathbf{b} = s_1\mathbf{a}_1 + \cdots + s_n\mathbf{a}_n$.

*Proof.* By Theorem 0.9.3, $\mathbf{As} = s_1\mathbf{a}_1 + \cdots + s_n\mathbf{a}_n$ for every $\mathbf{s} \in F^n$. Hence $\mathbf{s}$ is a solution, that is $\mathbf{As} = \mathbf{b}$, exactly when $\mathbf{b} = s_1\mathbf{a}_1 + \cdots + s_n\mathbf{a}_n$. This proves the second statement.

($\Rightarrow$) If the system is consistent, it has a solution $\mathbf{s}$, and then $\mathbf{b} = s_1\mathbf{a}_1 + \cdots + s_n\mathbf{a}_n$ is a linear combination of the columns. By Definition 1.3.2, $\mathbf{b} \in \operatorname{span}(\mathbf{a}_1, \ldots, \mathbf{a}_n)$.

($\Leftarrow$) If $\mathbf{b} \in \operatorname{span}(\mathbf{a}_1, \ldots, \mathbf{a}_n)$, then $\mathbf{b} = s_1\mathbf{a}_1 + \cdots + s_n\mathbf{a}_n$ for some $s_1, \ldots, s_n \in F$, and $\mathbf{s} = (s_1, \ldots, s_n)$ is a solution by the first paragraph. Hence the system is consistent.

■

Read backwards, the theorem is a recipe for Chapter 1 questions: to decide whether $\mathbf{b}$ lies in the span of some vectors of $F^m$, put the vectors as the columns of a matrix and ask whether the system is consistent.

**Example. Span membership as a system**

In $\mathbb{R}^3$, let $\mathbf{v}_1 = (1, 2, 3)$ and $\mathbf{v}_2 = (0, 1, 2)$. Decide whether $(1, 4, 7)$ and $(1, 4, 8)$ lie in $\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2)$.

*Solution.* Let $\mathbf{A}$ be the $3 \times 2$ matrix with columns $\mathbf{v}_1, \mathbf{v}_2$. By Theorem 3.1.1, $(1, 4, 7) \in$

$\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2)$ if and only if the system

$$a = 1, \qquad 2a + b = 4, \qquad 3a + 2b = 7$$

in the unknowns $a, b$ is consistent. The first equation gives $a = 1$, and then the second gives $b = 2$. The third equation must also hold, and it does: $3 + 4 = 7$. Hence $(1, 4, 7) = \mathbf{v}_1 + 2\mathbf{v}_2$ lies in the span.

For $(1, 4, 8)$, the first two equations are the same, so any solution has $a = 1$ and $b = 2$. The third equation now demands $3 + 4 = 8$, which is false. The system is inconsistent, so $(1, 4, 8) \notin \operatorname{span}(\mathbf{v}_1, \mathbf{v}_2)$.

Geometrically, $\operatorname{span}(\mathbf{v}_1, \mathbf{v}_2)$ is a plane through $\mathbf{0}$ in $\mathbb{R}^3$. The first right-hand side lies on that plane and the second does not.

## 3.1.4 The shape of the solution set

We now describe the solution set of an arbitrary system, starting with the homogeneous case. Chapter 1 already did the work there: in Example (Solutions of a Homogeneous System) we checked, with the three conditions of the subspace test, that the solutions of $\mathbf{Ax} = \mathbf{0}$ form a subspace.

**Theorem 3.1.2. Homogeneous solutions form a subspace**

Let $\mathbf{A} \in M_{m\times n}(F)$. The solution set $N = \{\mathbf{x} \in F^n : \mathbf{Ax} = \mathbf{0}\}$ of the homogeneous system $\mathbf{Ax} = \mathbf{0}$ is a subspace of $F^n$.

*Proof.* This is Example (Solutions of a Homogeneous System): $\mathbf{A0} = \mathbf{0}$, and for $\mathbf{x}, \mathbf{y} \in N$ and $c \in F$, the rules of Theorem 0.9.4 give $\mathbf{A}(\mathbf{x} + \mathbf{y}) = \mathbf{Ax} + \mathbf{Ay} = \mathbf{0}$ and $\mathbf{A}(c\mathbf{x}) = c(\mathbf{Ax}) = \mathbf{0}$. By the subspace test (Theorem 1.2.1), $N$ is a subspace of $F^n$. ∎

The solutions of $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ behave differently.

*Warning.* **The solution set of $\mathbf{Ax} = \mathbf{b}$ with $\mathbf{b} \neq \mathbf{0}$ is never a subspace.** A subspace must contain $\mathbf{0}$, but $\mathbf{A0} = \mathbf{0} \neq \mathbf{b}$, so $\mathbf{0}$ is not a solution. For example, the solutions of $x - y = 1$ in $\mathbb{R}^2$ form a line that misses the origin. The sum of the solutions $(1, 0)$ and $(2, 1)$ is $(3, 1)$, and $3 - 1 = 2 \neq 1$: adding two solutions does not give a solution.

What does survive is this: the **difference** of two solutions of $\mathbf{Ax} = \mathbf{b}$ solves the homogeneous system, since $\mathbf{A}(\mathbf{s} - \mathbf{s}') = \mathbf{b} - \mathbf{b} = \mathbf{0}$. So once one solution is known, every other one differs from it by a homogeneous solution. That is the whole structure.

**Theorem 3.1.3. Structure of the solution set**

Let $\mathbf{A} \in M_{m\times n}(F)$ and $\mathbf{b} \in F^m$, let $N$ be the solution set of $\mathbf{Ax} = \mathbf{0}$, and suppose $\mathbf{p} \in F^n$

is **one** solution of $\mathbf{Ax} = \mathbf{b}$. Then the solution set of $\mathbf{Ax} = \mathbf{b}$ is

$$\mathbf{p} + N = \{\mathbf{p} + \mathbf{h} : \mathbf{h} \in N\}.$$

*Idea.* The theorem is "particular plus homogeneous". To prove equality of two sets, prove both inclusions. Going from a solution $\mathbf{s}$ to $\mathbf{p} + N$ uses the difference $\mathbf{s} - \mathbf{p}$, as observed above; going back uses the sum $\mathbf{A}(\mathbf{p} + \mathbf{h}) = \mathbf{Ap} + \mathbf{Ah}$.

*Proof.* Let $S$ be the solution set of $\mathbf{Ax} = \mathbf{b}$. We use Theorem 0.3.1.

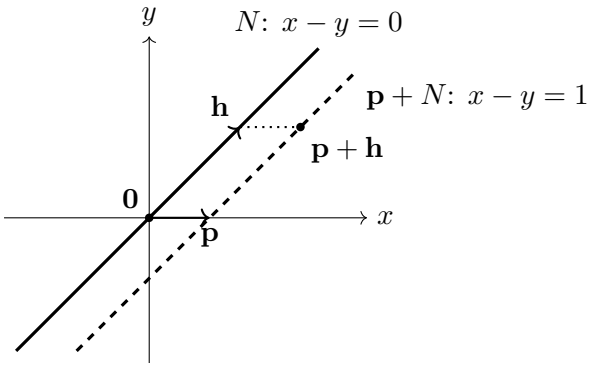
($\subseteq$) Let $\mathbf{s} \in S$, and put $\mathbf{h} = \mathbf{s} - \mathbf{p}$. By distributivity (Theorem 0.9.4), $\mathbf{Ah} = \mathbf{As} - \mathbf{Ap} = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so $\mathbf{h} \in N$. Hence $\mathbf{s} = \mathbf{p} + \mathbf{h} \in \mathbf{p} + N$.

($\supseteq$) Let $\mathbf{h} \in N$. Then $\mathbf{A}(\mathbf{p} + \mathbf{h}) = \mathbf{Ap} + \mathbf{Ah} = \mathbf{b} + \mathbf{0} = \mathbf{b}$, again by distributivity, so $\mathbf{p} + \mathbf{h} \in S$.

Therefore $S = \mathbf{p} + N$, as claimed.

■

The theorem splits solving $\mathbf{Ax} = \mathbf{b}$ into two independent jobs: find **one** solution, and find **all** solutions of the homogeneous system. The set $\mathbf{p} + N$ is a translate of the subspace $N$ by the vector $\mathbf{p}$, the kind of set that the picture in Example (A Line Missing the Origin) showed. For $x - y = 1$ in $\mathbb{R}^2$, the homogeneous solutions form the line $N = \text{span}((1, 1))$ through $\mathbf{0}$, one solution is $\mathbf{p} = (1, 0)$, and the solution set is the parallel line through $\mathbf{p}$.




A system solved by substitution fits the same pattern once we collect the free parameter.

**Example. Particular plus homogeneous**

Solve $x_1 + 2x_2 - x_3 = 4$, $x_2 + x_3 = 1$ over $\mathbb{R}$, and write the solution set in the form $\mathbf{p} + \text{span}(\mathbf{h})$.

*Solution.* The second equation gives $x_2 = 1 - x_3$. Substituting into the first, $x_1 = 4 - 2(1 - x_3) + x_3 = 2 + 3x_3$. The value of $x_3$ is not constrained, so write $x_3 = t$. Conversely, for every $t \in \mathbb{R}$ the vector $(2 + 3t, 1 - t, t)$ satisfies both equations, since $(2 + 3t) + 2(1 - t) - t = 4$ and $(1 - t) + t = 1$. Hence the solution set is

$$\{(2 + 3t, 1 - t, t) : t \in \mathbb{R}\} = \{(2, 1, 0) + t(3, -1, 1) : t \in \mathbb{R}\} = \mathbf{p} + \text{span}(\mathbf{h}),$$

with $\mathbf{p} = (2, 1, 0)$ and $\mathbf{h} = (3, -1, 1)$. As Theorem 3.1.3 predicts, $\mathbf{p}$ is a solution (take $t = 0$), and $\mathbf{h}$ solves the homogeneous system: $3 - 2 - 1 = 0$ and $-1 + 1 = 0$. In the row picture, two planes in $\mathbb{R}^3$ meet in a line, and this line does not pass through $\mathbf{0}$.

## 3.1.5 When is the solution unique?

A consistent system has a unique solution exactly when there is nothing to add to the particular solution. The theorem just proved turns this into a statement about the homogeneous system, and the column picture turns that into independence.

**Corollary 3.1.4. Uniqueness of solutions**

Let $\mathbf{A} \in M_{m\times n}(F)$ have columns $\mathbf{a}_1, \dots, \mathbf{a}_n$, and let $\mathbf{b} \in F^m$ be such that $\mathbf{Ax} = \mathbf{b}$ is **consistent**. The following are equivalent:

(a) $\mathbf{Ax} = \mathbf{b}$ has exactly one solution;

(b) $\mathbf{Ax} = \mathbf{0}$ has only the trivial solution;

(c) the list $(\mathbf{a}_1, \dots, \mathbf{a}_n)$ is linearly independent.

*Proof.* Let $N$ be the solution set of $\mathbf{Ax} = \mathbf{0}$. Since the system is consistent, it has a solution $\mathbf{p}$, and by Theorem 3.1.3 its solution set is $\mathbf{p} + N$.

(a $\Leftrightarrow$ b) The function $N \to \mathbf{p} + N$, $\mathbf{h} \mapsto \mathbf{p} + \mathbf{h}$, is surjective by definition of $\mathbf{p} + N$, and injective because $\mathbf{p} + \mathbf{h} = \mathbf{p} + \mathbf{h}'$ implies $\mathbf{h} = \mathbf{h}'$ by adding $-\mathbf{p}$ to both sides. So it is a bijection, and $\mathbf{p} + N$ has exactly one element if and only if $N$ does. Since $\mathbf{0} \in N$ always, $N$ has exactly one element if and only if $N = \{\mathbf{0}\}$, which is (b).

(b $\Leftrightarrow$ c) By Theorem 0.9.3, $\mathbf{Ax} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$ for every $\mathbf{x} \in F^n$. So (b) says: for all $x_1, \dots, x_n \in F$, $x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n = \mathbf{0}$ implies $x_1 = \cdots = x_n = 0$. This is word for word Definition 1.4.1 for the list $(\mathbf{a}_1, \dots, \mathbf{a}_n)$.

■

Two remarks on the hypotheses. The equivalence of (b) and (c) does not need consistency; only (a) does. And consistency cannot be dropped from (a $\Leftrightarrow$ b): the system $x = 0$, $x = 1$ in one unknown has only the trivial solution in its homogeneous version ($x = 0$, $x = 0$), but it has no solution at all, so it certainly does not have exactly one.

So the three pictures answer the two basic questions about $\mathbf{Ax} = \mathbf{b}$ in the language of Chapter 1: a solution **exists** when $\mathbf{b}$ is in the span of the columns, and it is **unique** when the columns are independent. What we still lack is a reliable way to decide either question for a given matrix. That is the job of Gaussian elimination, in the next section.

## 3.1.6 Exercises

### 3.1.6.1 A. Check your understanding

**Exercise. A1**

(a) Define what it means for a linear system $\mathbf{Ax} = \mathbf{b}$ over $F$ to be consistent, and what its augmented matrix is. What is the size of the augmented matrix of a system of $m$ equations in $n$ unknowns?

(b) Determine whether the following statement is true: "the solution set of $\mathbf{Ax} = \mathbf{b}$ is a subspace of $F^n$." Justify your answer.

(c) Determine whether the following statement is true: "if $\mathbf{Ax} = \mathbf{b}$ has two different solutions, then $\mathbf{Ax} = \mathbf{0}$ has a nontrivial solution." Justify your answer.

(d) Let $\mathbf{A} \in M_{3\times 5}(\mathbb{R})$. In which spaces do the unknown vector $\mathbf{x}$ and the right-hand side $\mathbf{b}$ of $\mathbf{Ax} = \mathbf{b}$ live?

(e) Name the theorem that turns "is $\mathbf{b}$ in the span of $\mathbf{a}_1, \dots, \mathbf{a}_n$?" into a question about a linear system, and state which system.

*Solution.*

(a) It is consistent if it has at least one solution $\mathbf{s} \in F^n$, that is, some $\mathbf{s}$ with $\mathbf{As} = \mathbf{b}$. The augmented matrix is $[\,\mathbf{A} \mid \mathbf{b}\,]$, the matrix $\mathbf{A}$ with $\mathbf{b}$ appended as an extra last column. It has size $m \times (n+1)$.

(b) False in general. If $\mathbf{b} \neq \mathbf{0}$, then $\mathbf{A0} = \mathbf{0} \neq \mathbf{b}$, so $\mathbf{0}$ is not a solution and the solution set is not a subspace. (If $\mathbf{b} = \mathbf{0}$, it is a subspace, by Theorem 3.1.2.)

(c) True. If $\mathbf{s} \neq \mathbf{s}'$ are solutions, then $\mathbf{A}(\mathbf{s}-\mathbf{s}') = \mathbf{b}-\mathbf{b} = \mathbf{0}$ by distributivity, and $\mathbf{s}-\mathbf{s}' \neq \mathbf{0}$.

(d) $\mathbf{x} \in \mathbb{R}^5$, one entry per column of $\mathbf{A}$, and $\mathbf{b} \in \mathbb{R}^3$, one entry per row.

(e) Theorem 3.1.1: $\mathbf{b} \in \operatorname{span}(\mathbf{a}_1, \dots, \mathbf{a}_n)$ if and only if $\mathbf{Ax} = \mathbf{b}$ is consistent, where $\mathbf{A}$ is the matrix with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$.

### 3.1.6.2 B. Practice

**Exercise. B1: Three views**

(a) Write the system $x_1 - x_2 + 2x_3 = 3$, $2x_1 + x_3 = 1$ over $\mathbb{R}$ in the matrix picture and in the column picture, and give its augmented matrix.

(b) Write out, as a list of equations, the system with augmented matrix $\left[\begin{array}{ccc|c} 1 & 4 & & 0 \\ 0 & 1 & & 2 \\ 3 & -1 & & 5 \end{array}\right]$ over $\mathbb{R}$. How many equations and unknowns does it have? Hence decide whether it is consistent.

*Solution.*

(a) The matrix picture is

$$\begin{pmatrix} 1 & -1 & 2 \\ 2 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}.$$

The column picture is $x_1(1,2) + x_2(-1,0) + x_3(2,1) = (3,1)$. The augmented matrix is $\left[\begin{array}{ccc|c} 1 & -1 & 2 & 3 \\ 2 & 0 & 1 & 1 \end{array}\right]$.

(b) The system is $x_1 + 4x_2 = 0$, $x_2 = 2$, $3x_1 - x_2 = 5$: three equations in two unknowns.

The second equation gives $x_2 = 2$, and then the first gives $x_1 = -8$. The third equation would require $-24 - 2 = 5$, which is false. Hence no vector satisfies all three equations, and the system is inconsistent.

**Exercise. B2: In the span or not?**

Determine whether the vector $\mathbf{b}$ lies in the span of the given vectors. Justify your answer, and when it does, write $\mathbf{b}$ as a linear combination of them.

(a) $\mathbf{b} = (2, 1, 4)$ and $(1, 0, 1), (0, 1, 2)$ in $\mathbb{R}^3$.

(b) $\mathbf{b} = (1, 1, 1)$ and $(1, 2, 0), (2, 4, 1)$ in $\mathbb{R}^3$.

(c) $\mathbf{b} = (1, 1, 0)$ and $(1, 0, 1), (0, 1, 1)$ in $\mathbb{F}_2^3$. Is the answer the same in $\mathbb{R}^3$?

*Solution.* By Theorem 3.1.1, in each case we decide whether the system $a\mathbf{v}_1 + b\mathbf{v}_2 = \mathbf{b}$ in the unknowns $a, b$ is consistent.

(a) The system is $a = 2$, $b = 1$, $a + 2b = 4$. The first two equations force $a = 2$, $b = 1$, and then $a + 2b = 4$ holds. Hence $(2, 1, 4) = 2(1, 0, 1) + (0, 1, 2)$ is in the span.

(b) The system is $a + 2b = 1$, $2a + 4b = 1$, $b = 1$. Doubling the first equation gives $2a + 4b = 2$, contradicting the second, since $2 \neq 1$. Hence the system is inconsistent and $(1, 1, 1)$ is not in the span.

(c) The system is $a = 1$, $b = 1$, $a + b = 0$. The first two force $a = b = 1$, and then $a + b = 1 + 1 = 0$ in $\mathbb{F}_2$. Hence $(1, 1, 0) = (1, 0, 1) + (0, 1, 1)$ is in the span over $\mathbb{F}_2$. In $\mathbb{R}^3$ the same first two equations force $a = b = 1$, but $1 + 1 = 2 \neq 0$, so the system is inconsistent and $\mathbf{b}$ is **not** in the span. The answer depends on the field.

**Exercise. B3: Solve and describe**

Solve each system over $\mathbb{R}$ by substitution, and write its solution set in the form $\mathbf{p} + \text{span}(\dots)$. Check that your $\mathbf{p}$ is a solution and that your spanning vectors solve the associated homogeneous system.

(a) $x_1 + x_2 + x_3 = 2$, $x_2 - x_3 = 1$.

(b) $x_1 - 2x_2 + x_3 - x_4 = 1$, $x_3 + 2x_4 = 3$.

*Solution.*

(a) The second equation gives $x_2 = 1 + x_3$, and then the first gives $x_1 = 2 - (1 + x_3) - x_3 = 1 - 2x_3$. With $x_3 = t$ free, every solution is $(1 - 2t, 1 + t, t)$, and conversely each such vector satisfies both equations. So the solution set is $(1, 1, 0) + \text{span}((-2, 1, 1))$. Check: $1 + 1 + 0 = 2$ and $1 - 0 = 1$; and $-2 + 1 + 1 = 0$, $1 - 1 = 0$.

(b) The second equation gives $x_3 = 3 - 2x_4$. The first then gives $x_1 = 1 + 2x_2 - (3 - 2x_4) + x_4 = -2 + 2x_2 + 3x_4$. The unknowns $x_2 = s$ and $x_4 = t$ are free, and every solution is

$$(-2 + 2s + 3t,\ s,\ 3 - 2t,\ t) = (-2, 0, 3, 0) + s(2, 1, 0, 0) + t(3, 0, -2, 1),$$

and conversely each such vector is a solution. So the solution set is $(-2, 0, 3, 0) +$

$\operatorname{span}((2,1,0,0),(3,0,-2,1))$. Check: $-2-0+3-0=1$ and $3+0=3$; for $(2,1,0,0)$: $2-2+0-0=0$ and $0+0=0$; for $(3,0,-2,1)$: $3-0-2-1=0$ and $-2+2=0$.

#### 3.1.6.3 C. Going deeper

**Exercise. C1: How many solutions?**

Let $\mathbf{A} \in M_{m\times n}(F)$ and $\mathbf{b} \in F^m$, and let $N$ be the solution set of $\mathbf{Ax}=\mathbf{0}$.

(a) Suppose $F$ is **infinite**. Prove that $\mathbf{Ax}=\mathbf{b}$ has either no solution, exactly one solution, or infinitely many solutions.

(b) Suppose $F$ is **finite**. Prove that $\mathbf{Ax}=\mathbf{b}$ has either no solution or exactly $|N|$ solutions.

(c) Give a system over $\mathbb{F}_3$ with exactly three solutions.

*Hint: for (a), if $N$ contains a nonzero vector $\mathbf{h}$, look at its multiples.*

*Solution.*

(a) Suppose the system is consistent, with a solution $\mathbf{p}$. By Theorem 3.1.3 its solution set is $\mathbf{p}+N$, and as in the proof of Corollary 3.1.4 the map $\mathbf{h}\mapsto\mathbf{p}+\mathbf{h}$ is a bijection $N\to\mathbf{p}+N$. If $N=\{\mathbf{0}\}$, there is exactly one solution. Otherwise, let $\mathbf{h}\in N$ with $\mathbf{h}\neq\mathbf{0}$. Since $N$ is a subspace (Theorem 3.1.2), $c\mathbf{h}\in N$ for every $c\in F$. If $c\mathbf{h}=c'\mathbf{h}$, then $(c-c')\mathbf{h}=\mathbf{0}$, and since $\mathbf{h}\neq\mathbf{0}$, Theorem 1.1.9 gives $c=c'$. So $c\mapsto c\mathbf{h}$ is injective, and $N$ contains infinitely many vectors because $F$ is infinite. Hence $\mathbf{p}+N$ is infinite.

(b) If the system is consistent with solution $\mathbf{p}$, the bijection $N\to\mathbf{p}+N$ from (a) shows that the solution set has exactly $|N|$ elements. Otherwise it has none.

(c) The single equation $x+y=1$ over $\mathbb{F}_3$ (a system with $m=1$, $n=2$). For each of the three values of $y\in\mathbb{F}_3$ there is exactly one $x=1-y$, so the solutions are $(1,0),(0,1),(2,2)$. Consistently with (b), the homogeneous equation $x+y=0$ has the three solutions $(0,0),(2,1),(1,2)$.

**Exercise. C2: The right-hand sides that work**

Let $\mathbf{A}\in M_{m\times n}(F)$ have columns $\mathbf{a}_1,\dots,\mathbf{a}_n$.

(a) Prove that the set $C=\{\mathbf{b}\in F^m : \mathbf{Ax}=\mathbf{b} \text{ is consistent}\}$ is a subspace of $F^m$.

(b) Deduce that if $n<m$, then there is some $\mathbf{b}\in F^m$ for which $\mathbf{Ax}=\mathbf{b}$ is inconsistent.

(c) For $\mathbf{A}=\begin{pmatrix}1&2\\2&4\\0&1\end{pmatrix}$ over $\mathbb{R}$, find one such $\mathbf{b}$.

*Solution.*

(a) By Theorem 3.1.1, $\mathbf{b}\in C$ if and only if $\mathbf{b}\in\operatorname{span}(\mathbf{a}_1,\dots,\mathbf{a}_n)$. So $C=\operatorname{span}(\mathbf{a}_1,\dots,\mathbf{a}_n)$, which is a subspace of $F^m$ by Theorem 1.3.1.

(b) By (a), $C$ is spanned by the $n$ vectors $\mathbf{a}_1,\dots,\mathbf{a}_n$. Since $\dim F^m=m$ (Example (Dimensions of the Running Examples)), Theorem 1.6.3 says no list of fewer than $m$

vectors spans $F^m$. As $n < m$, $C \neq F^m$, so some $\mathbf{b} \in F^m$ is not in $C$, and for that $\mathbf{b}$ the system is inconsistent.

(c) Take $\mathbf{b} = (1, 0, 0)$. A solution would satisfy $x + 2y = 1$ and $2x + 4y = 0$. Doubling the first gives $2x + 4y = 2 \neq 0$, a contradiction. So $\mathbf{Ax} = (1, 0, 0)$ is inconsistent.